Mayur Parvatikar

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EDUCATION

R. V. College of Engineering Bengaluru, India
Bachelor of Engineering in Information Science and Engineering *Aug. 2025*
CGPA: 9.28/10.0
Selected Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Artificial Intelligence and Machine Learning, Deep Learning, Natural Language Processing

SKILLS

Programming Languages: Python, Java, C++, SQL, JavaScript
ML/DL: PyTorch, TensorFlow, Hugging Face, scikit-learn, NumPy, Pandas, vLLM, XGrammar
LLM Methods: LoRA, DPO, SFT, Sparse Autoencoders, RAG, FAISS, LangChain
Computer Vision: OpenCV, U-Net, Vision Transformers, EasyOCR, SuryaOCR, TorchScript, Roboflow
Infrastructure: Docker, Kubernetes, AWS, Azure, GCP, Linux, Git, Jenkins CI/CD, Dynatrace
Backend: Spring Boot, Node.js, FastAPI, Redis

RESEARCH EXPERIENCE

Xu Lab, Carnegie Mellon University Pittsburgh, PA | Remote
Research Collaborator *Sept. 2025 – Present*

- Spearheaded **PATH-DISSECT**, applying mechanistic-interpretability tools to interpret what pathology foundation models represent and what their downstream classifiers rely on.
- Trained sparse autoencoders on frozen **UNI** and **Virchow2** foundation-model embeddings, decomposing dense, entangled features into sparse latents that align with pathologist-recognizable concepts.
- Demonstrated that the framework separates concept representation from classifier reliance and mitigates shortcut reliance in practice, accepted as a first-author paper at the **MI4MedFM** workshop at **MICCAI 2026**.

Pragya Lab, Birla Institute of Technology and Science, Pilani – Goa Campus Goa, India | Remote
Research Collaborator *Mar. 2025 – July 2026*

- Developed **Swiss-Knife**, a decode-time alignment framework that steers a frozen Qwen2.5-7B with hot-swappable helpful, honest and harmless DPO-LoRA adapters and switches objectives in **0.05 ms** without retraining.
- Achieved a harmonic **F1 of 0.797 vs. 0.750** over 5 decode-time baselines on the helpful-honest-harmless frontier, with the lowest refusal rate and **47% lower toxicity**, submitted to **AAAI 2027**.
- Built hidden-state extraction, torsion metrics and CKA, RepE and cosine baselines for **MENTIS** across 4 aligned 7–8B model pairs, where torsion separated safe from unsafe prompts at **AUC 0.89 vs. 0.64**, submitted to **AAAI 2027**.

R. V. College of Engineering Bengaluru, India
Undergraduate Researcher *July 2024 – Dec. 2024*

- Developed a user-aware recommendation system that maps a listener's emotion and health condition to therapeutic attributes of Indian classical music.
- Trained an LSTM-based speech emotion recognition model on the **RAVDESS** dataset using MFCC, RMS and ZCR features, reaching **93.6%** validation accuracy across 8 emotions.
- Evaluated recommendations with a custom Mean Reciprocal Rank metric and with precision and diversity against Spotify's live queue, winning the **Best Paper Presentation Award** at IEEE RAAI 2025, NTU Singapore.

Samsung R&D Institute India – Bangalore Bengaluru, India
Research Intern, PRISM Program *May 2023 – Nov. 2023*

- Trained U-Net-based architectures to segment static regions in gameplay frames on a custom Roboflow-annotated dataset.
- Utilized the masks with image inpainting for distortion-free frame interpolation via optical-flow and kernel-based algorithms.
- Validated outputs with PSNR, SSIM, LPIPS and Beyond Compare, earning the **$500 Certificate of Excellence**, the top honor.

ACADEMIC AND RESEARCH PROJECTS

Independent Research Project Bengaluru, India
Constrained Decoding Can Reverse Which Inputs Look Uncertain *May 2026 – Aug. 2026*

- Built a custom decoding loop that captures logits before and after XGrammar JSON masking on Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct across 100 controlled prompts.
- Discovered that grammar masking reverses which inputs look uncertain, shifting ranking **AUROC from 0.13–0.18 to 0.61–0.75** while post-mask entropy still looks normal, so confidence monitors receive no warning.

- Demonstrated that the reversal depends on how surviving probability mass is distributed rather than how much is removed, establishing pre-mask entropy as the reliable signal in a sole-author paper accepted at **UncertaiNLP @ EMNLP 2026**.

SPIRE Lab, Indian Institute of Science Bengaluru, India
Formbot OCR Pipeline, Machine Learning Project Intern *Oct. 2024 – Jan. 2025*

- Evaluated CRNN-based EasyOCR and ViT-based SuryaOCR against hand-built ground truth on **200+** low-quality government forms for Formbot, a voice assistant that helps citizens complete forms, exposing high word and character error rates.
- Engineered a preprocessing and form-splitting pipeline that cut the average error rate by **40%**, including hallucinated text.

INDUSTRY EXPERIENCE

SAP Labs India Bengaluru, India
Associate Platform Engineer, Data Privacy Engineering *July 2025 – Present*

- Engineered a reverse-proxy layer with OAuth-to-custom-token conversion that fronts downstream data-privacy services consumed by **200+ stakeholder applications**.
- Reduced Consent Management Service unavailability by **80%** by diagnosing bottlenecks with Dynatrace and JMeter load tests, eliminating N+1 queries, adding database views to cap HikariCP connections and introducing Redis caching.
- Executed a **zero-downtime** migration of **100K+ records** across canary and production regions on AWS, Azure and GCP using idempotent scripts, Jenkins pipelines and asynchronous polling APIs.
- Enabled multiple lines of business to onboard their AI agents to the platform by resolving agent memory, provisioning and Kubernetes deployment issues, and customized the n8n operator to enforce role-based credential access.

Platform Engineer Intern *Jan. 2025 – July 2025*

- Engineered Node.js microservices that automate metering-data validation and alerting, with OAuth-based authentication, exponential-backoff retries and integration with an email notification service.
- Built a “Try-It-Out” feature for Java Spring Boot APIs, published it on SAP’s internal API Hub and presented it to leadership, and resolved a **critical** penetration-testing vulnerability with JSoup validation.

Samsung R&D Institute India – Bangalore Bengaluru, India
Machine Learning Engineer Intern *May 2024 – July 2024*

- Implemented a novel encoder–decoder architecture for view synthesis from in-the-wild 2D images, delivering a real-time 3D rendering proof of concept for on-device AI.
- Integrated TorchScript models into an Android app with an in-app Python runtime via Chaquopy, reducing latency by **12%** through asynchronous execution and optimized image preprocessing.

PUBLICATIONS

[1] **M. Parvatikar**. "Constrained Decoding Can Reverse Which Inputs Look Uncertain." The 3rd Workshop on Uncertainty-Aware NLP (UncertaiNLP) at EMNLP 2026. Accepted.

[2] **M. Parvatikar**, X. Li, M. Xu. "PATH-DISSECT: Interpreting Concept Representation and Causal Reliance in Pathology Foundation Models with Sparse Autoencoders." Mechanistic Interpretability for Medical Foundation Models (MI4MedFM) Workshop at MICCAI 2026, Strasbourg, France. Accepted.

[3] A. Karmakar, **M. Parvatikar**, S. Dhoot, A. Dhanda, A. Chadha, K. Wanaskar, V. Jain, A. Das. "Swiss-Knife: A Framework for Reconfigurable Externalised Multi-Objective Alignment at Decode Time." arXiv:2609.22226, 2026. Submitted to AAAI 2027.

[4] P. P. Saha, S. Raina, **M. Parvatikar**, A. Dhanda, V. Jain, A. Chadha, A. Das. "MENTIS: What Belief Changes Under Alignment? Measuring Multi-Scale Latent Torsion in Language Models." arXiv:2606.01060, 2026. Submitted to AAAI 2027.

[5] **M. Parvatikar**, M. Mukunda, J. Chikkarangaiah, Mamatha G. S., G. Shobha. "A User-Aware Indian Music Recommendation System." 2025 5th International Conference on Robotics, Automation, and Artificial Intelligence (IEEE RAAI 2025), Singapore.

[6] **M. G. Parvatikar**, S. Rao, P. M. J. Rao, B. M. Sagar. "Deep Learning Approaches to Simile Detection: Insights from BERT and LSTM." 2024 15th International Conference on Computing Communication and Networking Technologies (IEEE ICCCNT 2024), IIT Mandi, India.

HONORS AND LEADERSHIP

Amazon ML Summer School, selected in the top 5% of 70,000+ applicants 2024
Secretary, IEEE Student Branch, R. V. College of Engineering – Bengaluru, India Dec. 2023 – Nov. 2024